

# Teaching Plan Generation and Evaluation With GPT-4: Unleashing the Potential of LLM in Instructional Design

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**Abstract**—This study explores and analyzes the specific performance of large language models (LLMs) in instructional design, aiming to unveil their potential strengths and possible weaknesses. Recently, the influence of LLMs has gradually increased in multiple fields, yet exploratory research on their application in education remains relatively scarce. In response to this situation, our research, grounded in pedagogical content knowledge theory, initially formulated an instructional design framework based on mathematical problem chains and corresponding prompt instructions. Subsequently, a comprehensive tool for assessing LLM's instructional design capabilities was developed. Utilizing Generative Pretrained Transformer 4, a high school mathematics teaching plan dataset was generated. Finally, the performance of LLMs in instructional design was evaluated. The evaluation results revealed that the teaching plans generated by LLMs excel in setting instructional objectives, identifying teaching priorities, organizing problem chains and teaching activities, articulating subject content, and selecting methods and strategies. Particularly commendable performance was noted in the modules of statistics and functions. However, there is room for improvement in aspects related to mathematical culture and interdisciplinary assessment, as well as in the geometry and algebra modules. Lastly, this study proposes initiatives, such as LLM prompt-based teacher training and the integration of mathematics-focused LLMs. These suggestions aim to advance personalized instructional design and professional development of teachers, offering educators new insights into the in-depth application of LLMs.

**Index Terms**—Automated teaching plan generation, instructional design capabilities, large language models (LLMs), mathematical problem chains, pedagogical content knowledge (PCK).

## I. INTRODUCTION

ARTIFICIAL intelligence (AI), as a transformative technology, is continuously permeating and altering our society and daily lives. Natural language processing, a key area in modern AI, has undergone major breakthroughs with the recent advancements of large language models (LLMs) [1]. These deep-learning models demonstrate remarkable language generation capabilities and profound understanding of complex knowledge, presenting extensive applications across many fields [2]. Among them, education emerges as a particularly promising domain for the integration of LLMs [3]. Specifically, LLMs can address the diverse needs of students and assist teachers in classroom instruction by undertaking a wide range of instructional tasks. Their applications include providing support on knowledge-based question answering and adaptive learning guidance, developing efficient and accurate teaching plans, and generating various types of educational materials [4], including teaching manuals, practice problems, and essay prompts. Furthermore, LLMs can provide personalized teaching suggestions and activities tailored to specific subject matter, instructional objectives, and student feedback, aiding teachers in better adjusting and optimizing the teaching process and plans. Education, as the pivotal sector of societal development, is gradually embracing and integrating a series of innovations driven by LLMs. These innovations offer new possibilities for the transformation of teaching and learning paradigms, with the innovation and improvement of educational methods becoming a vital support for the advancement of education. In this context, the innovations brought by LLMs are opening up new opportunities in the field of education [5].

Instructional design plays a pivotal role in the teaching process by facilitating student learning and development. It involves crafting a systematic and comprehensive teaching plan by setting clear learning objectives, employing effective teaching strategies, and implementing rigorous assessment methods [6]. A teaching plan represents a detailed plan crafted by teachers based on the curriculum and teaching resources, encompassing teaching content, methods, media, student activities, and evaluation. It serves as a crucial education tool reflecting

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teachers' instructional design approach and teaching strategy [7]. Creating an excellent teaching plan is challenging and time-consuming. Specifically, teachers need to possess a profound knowledge base and extensive teaching experience to design problem-based tasks tailored to students' diverse characteristics and to provide personalized guidance to better prepare students for the challenges of the 21st century. The rational design of problem-solving tasks in the teaching plan is also critical to a teacher's work. It not only poses questions to students for a deeper understanding of their knowledge, but also provides opportunities for exploration, conjecture, and the posing of meaningful questions, further facilitating the achievement of instructional objectives [8]. The integration of LLMs presents a promising avenue to further enhance the efficiency and quality of teachers' instructional design. Generative Pretrained Transformer (GPT)-4, as the latest model [9] in the GPT series developed by OpenAI [11], has exhibited outstanding natural language understanding and generation capabilities, as is evidenced by exceptional performance in various assessments and testing tasks. By leveraging the LLM's ability to understand subject knowledge and generate human-like language, instructors can create personalized learning plans and instructional pathways based on knowledge content and learning objectives. However, the adoption of new technologies in education necessitates a thorough understanding and rigorous evaluation of these technologies. Therefore, assessing the capability of LLMs in instructional design is crucial for recognizing their potential and limitations.

This study adopts the perspective of teachers' instructional design and views LLM as a "teacher agent focusing on instructional design." With high school mathematics as an example, the performance and limitations of the GPT-4 model (currently the most sophisticated model) in instructional design are evaluated. Specifically, the aim is to explore whether LLMs can be used to assist real teachers to enhance the quality and efficiency of teaching plan design and to provide new perspectives for personalized instructional design. In addition, it is currently challenging to evaluate the instructional design capabilities of LLMs without a suitable evaluation tool and dataset. To address this limitation, this study generates a teaching plan dataset and establishes an LLM evaluation tool, addressing the following questions. 1) How does the LLM perform across various assessment dimensions in instructional design? 2) How does the instructional design capability of LLM vary across different knowledge modules?

The primary contributions of this study are threefold. First, utilizing the GPT-4 model, this research generated a comprehensive dataset of high school mathematics teaching plans with carefully designing prompts instructing the model to use mathematical problem chains and specifying formats for teaching plan generation. To further showcase the advantage of mathematical problem chains in teaching plan generation, one-third of the teaching lessons were randomly selected, and generated a baseline teaching plan dataset using prompts without mathematical problem chains. Second, drawing from the research on pedagogical content knowledge (PCK) theory, an assessment tool for evaluating the generated teaching plans was developed.

This tool considers a wide range of factors, including knowledge content, instructional objectives, teaching methods and strategies, problem and situational design, teaching activities, and scope relevance. This evaluation framework allows for a more comprehensive evaluation of the generated teaching plans' quality and, in turn, reflecting the performance of LLMs in instructional design. Finally, an evaluation of the teaching plan dataset generated by the GPT-4 model was conducted. Through the scoring by professional mathematics teachers, the study analyzed the performance of GPT-4 in instructional design and provided suggestions for the application of LLM in teacher instruction.

The rest of this article is organized as follows. Section II begins with a review of related work, including research on PCK theory and instructional design framework, assessment of instructional design capabilities, existing studies on the design of mathematical problems, and the application of LLMs in the field of education. Section III provides a detailed description on the design of the teaching plan structure based on mathematical problem chains, the design of prompt commands for the generation of teaching plan dataset, and the development of the instructional design capability assessment tool. Section IV reports the analysis of the evaluation results. Section V discusses these findings and the implications for the application of LLM in teacher instruction. Section VI presents research limitations. Finally, Section VII concludes this article.

## II. LITERATURE REVIEW

### A. PCK and Instructional Design Framework

Designing an effective mathematics lesson requires teachers to skillfully integrate subject knowledge and pedagogical knowledge (PK) while aligning them with learners' unique interests and abilities. Subsequently, this integration should be effectively contextualized within specific themes or problems and presented during classroom instruction. Researchers refer to this ability as PCK [10]. Subsequent researchers have incorporated the concept of "technology" into the PCK framework, laying the foundation for the technological pedagogical content knowledge (TPACK) theory [11]. Within this framework, CK refers to the knowledge of the subject matter to be learned or taught, such as concepts, theories, and ideas. PK pertains to the knowledge of teaching and learning methods, lesson preparation, and assessment techniques. PCK encompasses strategies and methods essential for teaching specific content while incorporating fundamental knowledge of teaching and learning, insight of potential student misconceptions, and expertise in evaluation and reporting methodologies. In mathematics education, researchers conceptualize PCK as a fusion of content in student knowledge, PK, and curricular knowledge. They further categorize CK into general CK (GCK) and specialized CK (SCK) [12]. GCK is recognized as knowledge transferable across various disciplines [13], [14], whereas SCK pertains to distinct expertise in key mathematical domains, such as algebra, functions, geometry, probability, and statistics [15], [16]. While mathematics curriculum standards delineate the primary topics of instruction, the challenges for educators to design engaging lessons persist. This necessitates not

only a profound reservoir of knowledge but, more crucially, the adept integration of subject knowledge with pedagogical skills.

PCK is regarded as a core element of effective mathematical instruction [17], with teachers' PCK manifesting in both preclass instructional design and classroom teaching [18]. A teaching plan is an external representation of a teacher's instructional design capabilities, revealing the teacher's thought and design process [19]. The difference in instructional design abilities between novice and expert teachers lies in their ability to transform their CK into various forms of teaching and to adjust in response to students' capabilities and learning context [20]. A teaching plan emerges from teacher's thoughtful considerations of several key elements: the progression of learning (learning analysis), the cognitive tasks expected of learners (cognitive task analysis), the nature of classroom activities (activity analysis), and the structure of content (content/theme analysis). This results in the design of appropriate instructional content and sequence [21]. Teachers need to engage in thorough instructional design before the course begins to ensure effective teaching. Based on curriculum standards or syllabi (such as mathematics content and student mathematics literacy objectives) [22], teachers prepare necessary teaching materials (such as syllabi, presentations, homework plans, and practice tests) and design various teaching activities (such as role-playing, individual study, group work and collaborative inquiry) [6], ultimately creating an exemplary teaching plan. The fundamental elements of instructional design include instructional objectives, teaching activities, teaching strategies, and tools and methods for assessing students [23]. Researchers have explored the alignment of various instructional design components (including instructional objectives, teaching processes, and teaching assessment) to evaluate preservice teachers' abilities to meet students' learning needs through instructional design [7]. Misalignment between the learning objectives set by teachers and the actual teaching activities and assessments can impede student learning and mastery of subject knowledge [24]. In addition, the use of diverse teaching strategies and methods enables the realization of varied teaching activities and instructional paradigms [25]. Research indicates that teachers do not always employ the same teaching methods even under similar circumstances. Rather, they flexibly alternate among different teaching strategies and methods [26]. This flexible approach can encourage students to grasp new concepts and gain new experiences, thereby fostering the development of mathematical thinking, promoting students' mathematics performance, and affecting their attitudes toward learning [27].

### B. Assessment of Instructional Design Capabilities

The capabilities of teachers to design effective instruction, in both technology-integrated and traditional classroom settings [28], represent an important competency related to the PCK framework. Specifically, prior research indicates close interrelationships between teachers' instructional design skills and the domains of PCK and TPACK [29], [30]. Existing literature frequently examines teachers' instructional design and teaching abilities from a quantitative perspective, decomposing

PCK and TPACK into constituent dimensions for measurement and analysis [31]. For instance, some researchers have proposed a model for assessing technology-oriented instructional design grounded in the TPACK framework [32], delineating indicators of teachers' design capabilities and measuring them with methods including standardized self-reports, rating scales, and open-ended questionnaires [33], [34], and [35]. Researchers have also attempted to develop instructional plan analysis tools [36], mapping TPACK onto various elements of instructional design. By analyzing the content of teaching plans, they assess the instructional design abilities of preservice and mathematics teachers [19], [37]. In addition, some researchers, integrating the revised Bloom's taxonomy, have developed the 2-D EFL-TPACK scale [38], breaking down PCK into evaluative dimensions, such as instructional design, teaching strategies, and learning assessment. Finally, some researchers have incorporated sociocultural factors (i.e., mathematical history and culture) [39] into the evaluation framework of instructional design capabilities. They found that cultural and historical narratives about specific mathematical topics can stimulate students' insightful understanding [40].

### C. Problem Chains and the Application of LLMs in Education

In classroom teaching, instructional activities often take the form of teacher–student dialogues [41], and the design of questions posed by teachers is a crucial component of instructional design. Effective question design includes framing initial questions, articulating follow-up questions, and facilitating transitions between ideas [42]. Researchers categorize textbook questions along three dimensions: nontraditional versus traditional questions, open-ended versus closed questions, and those applicable to the real world versus those without practical application [43]. In mathematics textbooks, the problems are independently designed according to different knowledge points and corresponding contexts. These problems have diverse contexts, making it challenging for teachers to coordinate these contexts within the same lesson. Such compartmentalization limits the direct use of textbook questions for instructional purposes. As a result, teachers must redesign textbook questions to increase their real-life fidelity, thereby enabling students to understand and apply mathematical knowledge in more meaningful ways [44]. The quality of question design reflects a teacher's understanding and application abilities of subject knowledge [45]. In instructional design, skillfully transitioning and connecting various knowledge points through problem chains or sequences [46] is a crucial skill that distinguishes novice teachers from experts [47] [48]. Research indicates that if the designed questions are closely interconnected, they can effectively enhance students' problem-solving abilities [49]. The mathematical contexts, initial questions, and a series of subquestions in a problem chain are not meant to replace textbook problems but to supplement the regular questions found in textbooks [42]. The way teachers design questions can serve as a measure of their understanding of subject content [50].

Researchers have attempted to automate question-and-answer activities in classroom teaching using computers. For example,

the development of intelligent tutoring systems and ask systems involves using human–computer dialogues and adaptive question generation [5] to provide contexts, problems, and hypermedia resources for the purpose of tutoring students [51]. This approach conceals the teacher’s instructional design process within the algorithmic “black box” of question generation and learning resource recommendation, offering consolidation exercises based on each knowledge point to help students master the content. However, in the learning process of each lesson for students, what questions should be posed at each stage and the roles these questions play are beyond the current capabilities of intelligent systems. This aspect of instructional design and analysis still requires the involvement of teachers [52]. Recent advances in LLMs, such as GPT-3 [1] and pathways language model (PaLM) [2], have demonstrated strong question-answering and reasoning abilities [53], [54]. Especially when paired with chain of thought prompts, these models exhibit aptitude for mathematical word problems [55], illuminating new possibilities for question design in mathematics instruction [56]. Researchers have utilized the GPT-3.5-turbo application programming interface (API) for question generation [57], and some have even attempted to develop interactive chat tools, such as GPTeach, to assist teachers in designing questions for interactions with GPT-simulated students [3]. Others have explored using LLMs to assist teachers in rapidly creating course outlines, although this process still requires human input and expertise. As tools for preparing CK and course outlines, LLMs might generate text that is irrelevant or inappropriate, subsequently lowering the quality of the course outline [58].

Current research suggests that LLMs hold promise for enhancing educational productivity, learning, innovation, and creativity [59]. They are transforming teachers’ daily instructional method by providing instant feedback and assistance, thus making the learning process more efficient [60]. However, researchers also point out that current LLMs are considerably inadequate as teachers, particularly in terms of response consistency and accuracy, and still lack the teaching skills necessary to qualify as competent educators [61]. Compared with broader curricula syllabi, individual teaching plans contain more specific instructional details, deeply reflecting a teacher’s experience and thought process, and also showcasing their instructional design capabilities. It is crucial to critically assess the teaching-related capabilities of LLMs and analyze their suitability for specific educational scenarios [62]. This study views LLMs as “teacher agents focused on instructional design” and analyzes the teaching plans they generate to evaluate their instructional design capabilities, which is vital for determining the depth of LLMs’ applications in the field of education.

### III. RESEARCH DESIGN

The research design of the present study involves three main components: constructing an instructional design framework based on mathematical problem chains, generating a high school mathematics teaching plan dataset, and establishing the evaluation dimensions for LLM’s instructional design capabilities (as shown in Fig. 1). Collectively, this research design facilitates

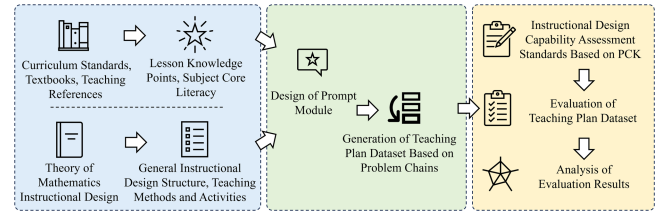


Fig. 1. Process of establishing the dataset and evaluation tools.

TABLE I  
INSTRUCTIONAL DESIGN FRAMEWORK BASED ON MATHEMATICAL PROBLEM CHAINS

| Component             | Phase                              | Elements                                                                                                                                                                                                                                           |
|-----------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching Analysis     | Instructional Objective            | Bloom’s classification of Instructional Objectives + Knowledge Points                                                                                                                                                                              |
|                       | Teaching Priorities and Challenges | Analysis of Knowledge Points and Learning Situation                                                                                                                                                                                                |
| Instructional Process | Pre-class Introduction             | Questions Introduced Through Designed Scenarios + Problem Chain 1 + Problem Chain 2 + Teaching Activities and Design Intentions                                                                                                                    |
|                       | Explanation of New Knowledge       | Knowledge Point 1 + Problem Chain 3 + Variant Problem 1 + Variant Problem 2 + Teaching Activities and Design Intentions<br>Knowledge Point 2 + Problem Chain 4 + Variant Problem 1 + Variant Problem 2 + Teaching Activities and Design Intentions |
|                       | Consolidation and Improvement      | Advanced Problem + Answer Hints + Advanced Exercise and Answer Hints 1 + Advanced Exercise and Answer Hints 2 + Teaching Activities and Design Intentions                                                                                          |
|                       | Comprehensive Exercise             | Comprehensive Exercises and Answer Hints + Variant Exercise and Answer Hints + Teaching Activities and Design Intentions                                                                                                                           |
|                       | Lesson Summary                     | Teacher Activity 1 + Teacher Activity 2 + Summary of Lesson Knowledge Content                                                                                                                                                                      |

an in-depth assessment of LLM’s performance in instructional design. The following sections provide detailed descriptions on each component.

#### A. Establishing an Instructional Design Framework Based on Mathematical Problem Chains

To objectively evaluate the instructional design capabilities of LLM, it is essential to establish a unified instructional design framework to ensure a consistent structure across LLM’s generated teaching plans. In addition, including output format requirements in prompt instructions can reduce the randomness of LLM-generated responses [59]. Since there is currently no instructional design framework based on mathematical problem chains, it is necessary to integrate problem chain design into the general instructional design structure and conduct a more detailed reconstruction.

The general instructional design framework includes components, such as teaching analysis, instructional process, classroom blackboard writing, teaching reflection, and postclass assignments. Specifically, teaching analysis focuses on creating instructional objectives, identifying teaching priorities, and addressing teaching challenges. The instructional process includes stages, such as preclass introduction, explanation of new knowledge, consolidation and improvement, comprehensive exercise, and lesson summary. Building upon this general structure, this study drew upon mathematics teaching theory and instructional design principles [63] to develop an instructional design framework centered on mathematical problem chains. This specially designed problem chain includes a sequence of elements, such as situational introduction to the problem, knowledge explanation questions, knowledge consolidation questions, and advanced

enhancement questions, as given in Table I. The preclass introduction phase incorporated hypothetical scenarios and problem chains to provide context and spark student interests. The explanation of the new knowledge phase built upon these scenarios and problem chains, adding question variants to expand conceptual understanding. To reinforce learning, the consolidation and improvement phase presented advanced problems related to the earlier scenarios and chains. The comprehensive exercise phase, unlike previous stages, focuses on providing students with practical experience applying mathematical concepts. Therefore, it does not build upon the original scenarios and problem chains but presents a wide variety of practice problems. At the end of each phase, the descriptions of teaching activities and the underlying design intentions were added. This approach allows the LLM, following the formulation of questions, to determine the appropriate teaching activities and methods, and to provide explanations for the chosen design intentions. This methodology aligns with current research that emphasizes the design of best-fitting teaching activities for various knowledge points, as well as the elaboration of underlying principles and strategies behind these designs [64]. Finally, in the lesson summary phase, a summary of the teacher's teaching activities and the knowledge content covered in the lesson was added, examining the LLM's accuracy in understanding the knowledge content and its ability to generalize. This design approach aligns more closely with actual teaching needs, thus forming an instructional design structure based on mathematical problem chains.

### B. Design of Prompt Instructions and Generation of Mathematics Teaching Plan Dataset

In this section, the present research translated the aforementioned instructional design framework based on mathematical problem chains into prompt instructions, commanding the LLM to generate corresponding teaching plan content. Analysis of existing research on LLM prompt engineering indicates that when sending prompt instructions to LLMs, more precise, standardized, and detailed descriptions can guide LLMs to generate higher quality responses, and zero-shot or few-shot prompting tends to yield better results [65]. Therefore, it is necessary to provide the model with detailed subject knowledge (CK) and PK to improve the accuracy and quality of the generated teaching plan. After preliminary testing and analysis of multiple Chinese language-supporting LLMs [66], it was found that the GPT-4 model performs best in terms of its ability to generate content and format for teaching plans. Other LLMs had issues, such as failing to follow the format requirements, being overly brief, or generating inaccurate information. Consequently, the GPT-4 model was selected for generating the teaching plan dataset.

First, based on the high school mathematics curriculum standards issued by China's Ministry of Education, mathematics textbooks, and the core competencies of the mathematics subject [67], this study organized and summarized the key knowledge points of each class period in high school mathematics. The high school mathematics curriculum, encompassing 21 chapters, was categorized into four main knowledge modules: Statistics,

TABLE II  
COMMON TYPES OF ELEMENTS IN HIGH SCHOOL MATHEMATICS  
INSTRUCTIONAL DESIGN

| Element             | Type                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lesson Introduction | Question-driven Introduction, Problem-driven Introduction, History of Mathematics Introduction, Math Story Introduction, Scenario Introduction, Media Introduction, Chart Introduction, Physical Object Introduction, Preview Introduction, Comparison Introduction, Music or Art Introduction                                                                                         |
| Teaching Methods    | Heuristic Teaching, Cooperative Learning, Situational Learning, Project-based Learning, Tiered Instruction, Blended Learning, Inquiry-based Teaching, Gamified Learning, Story-based Teaching, Direct Instruction                                                                                                                                                                      |
| Teaching Activities | Group Discussion, Guided Activities, Interactive Q&A, Online Learning, Brainstorming, Role-playing, Experimental Activities, Gaming Activities, Demonstration and Display, Field Trips, Classroom Debate, Creative Activities, Self-reflection, Note-taking Activities, Writing Stories or Diaries, Making Charts or Models, Reading Aloud, Knowledge Contests, Collaborative Learning |
| Lesson Summary      | Student Speech Summary, Group Summary Discussion, Mind Mapping, Quick Quiz                                                                                                                                                                                                                                                                                                             |

Functions, Algebra, and Geometry [15]. The specific chapters included in each knowledge module are as follows.

- 1) The "Statistics" module covers introductory and advanced probability, statistics, principles of counting, and statistical analysis of paired data.
- 2) The "Functions" module includes concepts, properties, and applications of functions, power functions, exponential and logarithmic functions, trigonometric functions, derivatives, and their applications.
- 3) In the "Algebra" module, topics include sets and logic, equations and inequalities, powers, exponents and logarithms, complex numbers, and sequences.
- 4) Finally, the "Geometry" module encompasses trigonometry, plane vectors, lines and planes in space, simple geometric solids, lines in the plane rectangular coordinate system, conic sections, spatial vectors, and their applications.

Second, the present study compiled a list of common teaching methods and activities used in high school mathematics through literature review and interviews with mathematics teachers [68]. Drawing upon the theory of mathematics instructional design, these elements were organized into four major categories (preclass introduction methods, teaching methods, teaching activities, and lesson summary), as outlined in Table II. First, 11 preclass introduction methods were identified, ranging from discussing the history of mathematics to using charts. These methods can diversify the teaching plan's introductory approach generated by the LLM. Second, ten distinct teaching methods (e.g., tiered teaching and inquiry-based teaching) were listed, and the unique advantages and suitable scenarios associated with each method were elaborated on. This can provide LLM with a broad spectrum of teaching strategy options. Third, the teaching activities category includes 19 diverse strategies, such as group discussions and interactive question-and-answer sessions. This list can guide the LLM to make flexible selections tailored to specific content and instructional objectives, enriching the content of generated teaching plans and increasing its validity. Finally, the lesson summary category includes five methods, such as teacher summarization and mind mapping. This category allows LLM to choose an appropriate method for lesson summary to increase the completeness and coherence of the teaching plan.

The content of these CK and PK elements was combined and sent to the LLM, instructing the LLM to select suitable methods and strategies to generate teaching plan content, thereby unleashing its full potential in instructional design and maximizing the quality of the generated content.

Finally, using a step-by-step inquiry approach, this study constructed prompt instructions for each teaching phase, combining zero-shot and few-shot prompts to generate teaching plan content. As given in Table III and the Appendix, the prompt instructions for each teaching phase include lesson knowledge points, output format, and transitional prompts that connect with the content of the previous phase. In the preclass introduction phase, the present research provided few-shot prompt examples for situational descriptions and problem chains. The prompts were composed in the format of “lesson name + instructional objective + knowledge points + preclass introduction method + example of situational introduction, and problem chain + output format.” Similarly, in the explanation of new knowledge phase, few-shot prompt examples for problem chains and variation questions were included, composed in the format of “lesson name + preclass introduction + knowledge points + teaching activity + teaching method + example of problem chain, and variation question + output format.” Through these standardized formats connecting each phase, the prompts guided the LLM to connect the knowledge content of the entire lesson into coherent teaching plans within the context of the problem chains, while designing the best-fitting variant questions and advanced questions. Finally, the output of all these inquiries were combined into a complete teaching plan. If the generated content did not meet the format requirements, the inquiry process was reinitiated. After over 1300 rounds of inquiry, a high school mathematics teaching plan dataset covering 21 chapters and 183 lessons was successfully generated, totaling 605 377 words of text content, providing ample material to evaluate LLM’s performance in instructional design. In addition, 15 lessons from each of the four knowledge modules were randomly selected, and conducted step-by-step inquiries of GPT-4 with zero-shot prompts containing only the lesson name and output format. The results of these inquiries were also compiled into a complete teaching plan, totaling 60 lessons, which served as the baseline teaching plan dataset generated by GPT-4. Meanwhile, the present study collected the teaching plans corresponding to these 60 lessons from the National Smart Education Public Service Platform in China. These outstanding teaching plans were crafted by exemplary teachers with over ten years of teaching experience, and used as a dataset of baseline teaching plans designed by human teachers for this research.

### C. Development of an Instructional Design Capability Assessment Tool

Currently, the evaluation of LLMs primarily focuses on assessing answer accuracy [69], while research on evaluating the quality of generated text content remains relatively scarce [55]. Therefore, to ensure the comprehensiveness and effectiveness of the assessment, it is necessary to establish evaluation metrics for assessing LLM’s performance in instructional design.

Referencing existing PCK and TPACK assessment scales, an evaluation tool was constructed, comprising dimensions, such as instructional objectives, analysis of learning situations and teaching materials, problem chains, teaching activities, knowledge content, teaching methods and strategies, teaching evaluation, interdisciplinary aspects, applicability, scope, and an overall rating. This tool includes 20 items across 11 dimensions, as given in Table IV. To achieve finer scoring granularity, an eight-point Likert scale was utilized, where 1 point represents “strongly disagree” and 8 points represent “strongly agree.” In addition, to better distinguish between excellent and poor teaching plan content and to avoid a large occurrence of median values in scoring, the present study did not set a middle score representing “average.”

To better address research questions, this research invited three certified high school mathematics teachers, each with at least five years of teaching experience, to evaluate and score the generated teaching plans. In addition, two certified mathematics teachers with over ten years of experience were asked to review content where there were significant discrepancies in scoring (refer to the Appendix of this article for basic information about the teachers). To ensure consistency in the assessment, a detailed explanation of each evaluation dimension and scoring examples were provided to the teachers. The study also introduced PCK and TPACK theories to ensure that each mathematics teacher understood and agreed with each evaluation dimension. During the evaluation process, each mathematics teacher spent at least 8 min reviewing each teaching plan. The interrater reliabilities were assessed by calculating the two-way random effect intraclass correlation coefficients with three raters (ICC [2, 3]). ICC (2, 3) was  $0.768 > 0.6$ , indicating good agreement in the teachers’ assessments of the generated teaching plans. For individual teaching plans, where significant scoring discrepancies for certain dimensions exist, the researchers of this study further explored the reasons behind the teachers’ scores to achieve as much consistency as possible. These scoring discrepancies and their reasons will be elaborated in detail in Section V of this article.

## IV. RESULTS ANALYSIS

In this section, the study analyzed the evaluation results of the teaching plan dataset to assess the performance of LLM in instructional design. The results in this section were divided into two sections: the first section involved analyzing the evaluation dimensions using descriptive statistical methods, and the other focused on cross-analysis between different knowledge modules and evaluation dimensions.

### A. Descriptive Statistical Analysis for Each Dimension

In this section, a descriptive statistical analysis of the 20 evaluation dimensions was conducted. The sample size for the teaching plan dataset based on mathematical problem chains was  $N=183$ . The baseline dataset of teaching plans generated by GPT-4 and the dataset of exemplary teaching plans designed by human teachers both had a sample size of  $N=60$  (see Fig. 2).

From the perspective of each dimension, the average score for the Q1 dimension (design of problem situational introduction)

TABLE III  
DESIGN OF PROMPT INSTRUCTIONS AND GENERATED TEACHING PLAN CONTENT EXAMPLES

| Phase                        | Prompt Design                                                                                                                                                     | LLM Q&A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Preclass Introduction        | Lesson Name + Instructional Objectives + Knowledge Points + Methods of Pre-Class Introduction + Examples of Case Introductions and Problem Chains + Output Format | <p><b>Question:</b><br/>You are currently designing the "Pre-class Introduction" in the teaching plan. Subsequently, there will be segments on "Explanation of New Knowledge," "Consolidation and Improvement," "Comprehensive Exercise," and "Lesson Summary" that you need to design. Integrate the previously generated "Instructional Objective" to analyze and design the "Pre class Introduction" teaching activity for the "Stratified Random Sampling" lesson in the "Statistics" unit. Ensure smooth transition and continuity with the segments that will be designed later. <i>The knowledge content to be taught in this lesson includes:</i> stratified random sampling or stratified sampling, and the methods of stratified sampling. <i>The "preclass introduction" can refer to the following methods:</i> inquiry introduction, problem introduction, history of mathematics introduction, math story introduction, scenario introduction, media introduction, chart introduction, real object introduction, preview introduction, comparison introduction, and music or art introduction. Please provide a detailed and specific teaching plan, divided into a case introduction and a problem chain based on the case introduction. Note that the case introduction and the problem chain should contain specific and detailed values and problems. Please do not exceed the content knowledge in your design. If calculations are needed, list the mathematical formulas and plug in the values using a calculation plugin. Below is an example of the "Pre-class Introduction":<br/>(<i>Case introduction</i>): In a right triangle, the sine, cosine, tangent, and cotangent of an acute angle have been defined. Which quantities in the right triangle are these four values related to?<br/>(<i>Problem Chain 1</i>): How can we define the sine, cosine, tangent, and cotangent of any angle using the Cartesian coordinate system?<br/>(<i>Problem Chain 2</i>): Does the sine, cosine, tangent, and cotangent of angle <math>\alpha</math> have meaning for any angle?<br/>(<i>Teaching activities and design intentions</i>): By creating relevant scenarios to arouse students' interest, new concepts are introduced in specific contexts, enabling students to better understand and accept definitions and formulas.</p> <p><b>Answer:</b><br/>(<i>Case introduction</i>): Suppose we are a market research company commissioned by a large supermarket to conduct a study on consumer purchasing habits. Our task is to sample the supermarket's customers, but the supermarket's customers are numerous and diverse. We could simply randomly select some people, but this may result in underrepresentation of certain groups, such as young people or the elderly. How can we design a more effective sampling scheme?<br/>(<i>Problem Chain 1</i>): For such a complex population, what issues might arise if we only use simple random sampling? Is it possible that some groups may be under-sampled?<br/>(<i>Problem Chain 2</i>): How can we avoid this issue in simple random sampling? Can we classify the population based on certain characteristics and then draw a proportional sample from each category?<br/>Does this grouped sampling method have a specific name? What are its main steps and methods?<br/>(<i>Teaching activities and design intentions</i>): By setting up a real-world scenario, students' interest in stratified random sampling is sparked. It allows students to ponder how to address practical challenges, naturally leading to the introduction of the concept of stratified random sampling. This approach enables students to better grasp the principles and applications of stratified random sampling, and they can effectively apply it to solve real-world problems in subsequent learning.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Explanation of New Knowledge | Lesson Name + Preclass Introduction + Knowledge Points + Teaching Activities + Teaching Methods + Examples of Problem Chains and Variant Problems + Output Format | <p><b>Question:</b><br/>Now, please design the content for "Explanation of New Knowledge" for the "Stratified Random Sampling" lesson in the "Statistics" unit, incorporating the context from the previously generated pre-class introduction.<br/><i>New knowledge to be taught in this lesson includes:</i> stratified random sampling or simply stratified sampling, and methods of stratified sampling. <i>The design for "Explanation of New Knowledge" can refer to the following learning activities:</i> group discussions, guided activities, interactive Q&amp;A, online learning, brainstorming, role-playing, experimental activities, game-based activities, demonstrations and presentations, field observations, classroom debates, creative activities, self-reflection, shorthand activities, writing stories or diaries, creating charts or models, reading aloud, knowledge contests, cooperative learning, etc. <i>Teaching methods to refer to include:</i> heuristic teaching, cooperative learning, situational learning, project-based learning, tiered instruction, blended learning, inquiry-based teaching, gamified learning, and storytelling teaching methods, etc. Please design detailed and specific activities for each knowledge point in the form of a detailed teaching plan. Ensure coherence with the pre-class introduction and the design of the problem chain. Generate variant questions based on the content of the pre-class introduction scenario, and present them in the form of problem chains and variant questions. After each question, provide possible teaching activities and design intentions. Please refrain from designing beyond the content scope. If calculations are needed, list the mathematical formulas and plug in the numbers using the relevant tools.<br/>Here is the content format:<br/><b>Knowledge point 1:</b><br/>(<i>Problem Chain 3</i>): Given that the terminal side of angle <math>\alpha</math> passes through point P(1,2), find the sine, cosine, tangent, and cotangent values of angle <math>\alpha</math>.<br/>(<i>Variant Question 1</i>): If the coordinates of point P in Problem Chain 3 are changed to P(a,-2a) (<math>a&gt;0</math>), will the sine, cosine, tangent, and cotangent values of angle <math>\alpha</math> change? Why?<br/>(<i>Variant Question 2</i>): Given that the terminal side of angle <math>\alpha</math> passes through point P(a,-2a) (<math>a&lt;0</math>), find the sine, cosine, tangent, and cotangent values of angle <math>\alpha</math>.<br/>(<i>Teaching Activities and Design Intentions</i>):<br/><b>Knowledge Point 2:</b><br/>(<i>Problem Chain 4</i>):<br/>(<i>Variant Question 1</i>):<br/>(<i>Variant Question 2</i>):<br/>(<i>Teaching Activities and Design Intentions</i>):</p> <p><b>Answer:</b><br/><b>Knowledge Point 1:</b> What is stratified random sampling?<br/>(<i>Problem Chain 4</i>): Based on the pre-class introduction, we have already categorized the supermarket's customer groups. Can such an operation solve the potential problems we discussed in simple random sampling?<br/>(<i>Variant Question 1</i>): Suppose we categorize based on two dimensions: gender and age, and randomly select a certain number of people from each category for the survey. Can this method better represent the overall customer group? What is this sampling method called?<br/>(<i>Teaching Activities and Design Intentions</i>): Interactive Q&amp;A activity, encouraging students to think and answer questions, guiding them to discover the concept of stratified random sampling on their own. The goal of the instructional design is to let students understand through this activity that stratified random sampling is a more effective sampling method that can better represent the entire research group.<br/>(<i>Knowledge Point 2</i>): How to carry out stratified random sampling?<br/>(<i>Problem Chain 5</i>): We already understand the basic concept of stratified random sampling. So, how do we proceed with stratified random sampling? If we categorize the supermarket's customers by gender and age, how many people should we select in each category?<br/>(<i>Variant Question 1</i>): Suppose we categorize male customers by age groups: 18-25 years, 26-35 years, 36-45 years, and over 46 years. Female customers are also categorized in the same age brackets. How many people should we select in each category? How do we decide?<br/>(<i>Teaching Activities and Design Intentions</i>): Group discussion activity, encouraging students to understand and grasp the specific methods of stratified random sampling through discussion and exploration. Through this activity, students understand the factors and steps to be considered in stratified random sampling, enhancing their logical reasoning and mathematical modeling abilities.</p> |

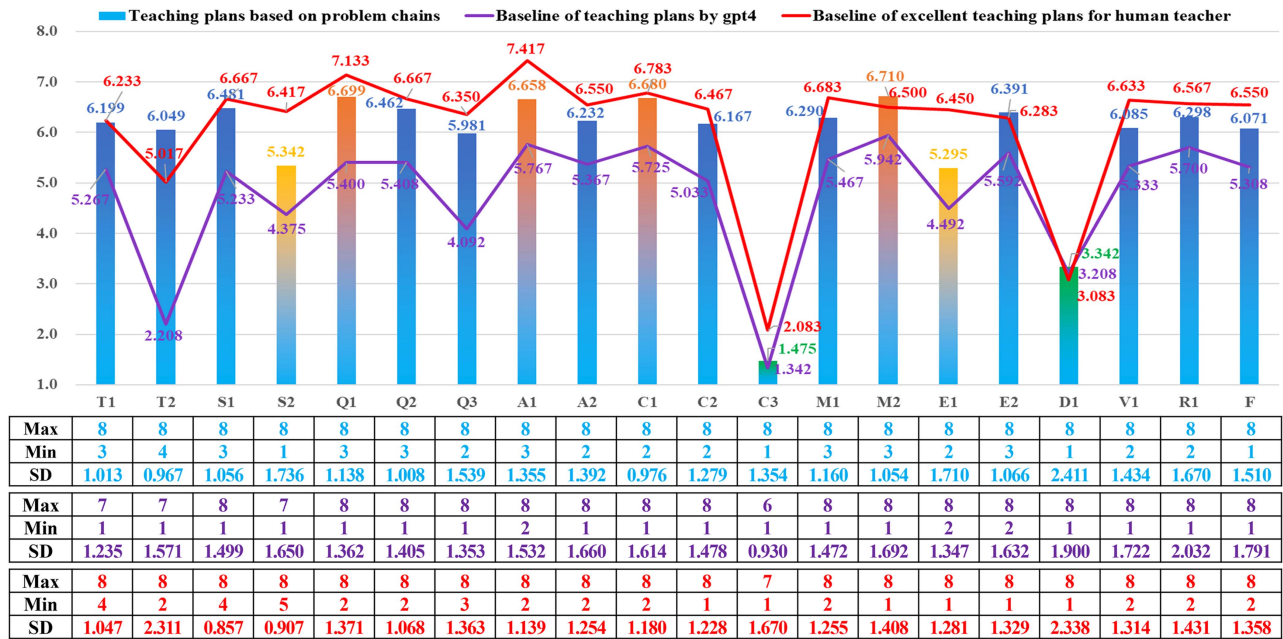


Fig. 2. Average values of assessment dimensions.

TABLE IV  
DIMENSIONS FOR ASSESSING LLM'S INSTRUCTIONAL DESIGN CAPABILITY

| Category                                 | Number | Dimension                                                                                                                         | Sources   |
|------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| Instructional Objectives                 | T1     | Ability to design instructional objectives that align with the class content and are logically formulated                         | [70] [28] |
|                                          | T2     | Ability to design instructional objectives that reflect the core literacy of the subject                                          | [70] [28] |
| Teaching Situation and Textbook Analysis | S1     | Recognize the emphasis points of the teaching content                                                                             | [28]      |
|                                          | S2     | Identify common misunderstandings and learning difficulties of students                                                           | [71]      |
| Problem Chain                            | Q1     | Design and introduce rich, effective, and scientific problem scenarios                                                            | [72]      |
|                                          | Q2     | Design challenging learning problems and tasks that connect and promote step by step                                              | [72]      |
|                                          | Q3     | Design suitable variant exercises to consolidate learned knowledge                                                                | [72]      |
| Teaching Activities                      | A1     | Design content that promotes teacher–student and student–student interactive discussions                                          | [71]      |
|                                          | A2     | Select teaching activities that align with the teaching objectives and content                                                    | [28]      |
| Knowledge Content                        | C1     | Correctly and reasonably explain the basic theories and concepts of the subject                                                   | [28] [72] |
|                                          | C2     | Extract and summarize the subject content of the teaching period                                                                  | [28]      |
|                                          | C3     | Introduce the historical development of important theories in the subject and the culture of mathematics                          | [28]      |
| Teaching Methods and Strategies          | M1     | Choose suitable teaching methods and strategies                                                                                   | [71]      |
|                                          | M2     | Apply a variety of teaching methods and strategies                                                                                | [28]      |
| Teaching Evaluation                      | E1     | Recognize the individual differences and learning differences among students                                                      | [73] [74] |
|                                          | E2     | Use a variety of ways to assess students' learning situation                                                                      | [73] [28] |
| Interdisciplinary                        | D1     | Establish appropriate connections with other disciplines and design activities that develop students' interdisciplinary abilities | [74] [71] |
| Usable Value                             | V1     | Can be used for secondary modification and to provide reference value for teaching plans                                          | \         |
| Scope                                    | R1     | Ability to design the content within the scope of the curriculum standard                                                         | [73]      |
| Overall Score                            | F      | Overall evaluation of the teaching plan                                                                                           | \         |

was 6.699, the A1 dimension (design of teacher–student and student–student interactive activities) was 6.658, C1 dimension (explanation of subject content) was 6.680, and M2 dimension (ability to use a variety of teaching methods and strategies) was 6.710, all these dimensions scored exceeding 6.5. Furthermore, the score for the T2 dimension (design of teaching objectives that reflect the core competencies of the subject) was 6.049, surpassing the baseline score of both GPT-4 generated and human-designed teaching plans. These findings revealed that LLM

performed exceptionally well in setting instructional objectives, designing teaching situations, selecting teaching activities and method strategies, and explaining subject knowledge concepts. However, the average score for the S2 dimension (identifying common student misconceptions and learning difficulties) was 5.342, and for the E1 dimension (analyzing individual and learning differences among students) was 5.295. This indicates that LLM performed moderately in analyzing teaching challenges, common student misconceptions, and designing tiered teaching activities. The average scores for the D1 dimension (interdisciplinary) and C3 dimension (mathematical culture) were 3.342 and 1.475, respectively. These scores were similar to the evaluation performance of both the baseline teaching plans generated by GPT-4 and those designed by human teachers. This suggests that LLM is still lacking in these areas; the content of the teaching plans shows insufficient interconnection with other subjects and fails to incorporate aspects of mathematical culture and history effectively.

It is noteworthy that for all 20 evaluation indicators, the teaching plans generated by GPT-4 based on mathematical problem chains performed better than those in the baseline teaching plan dataset generated by GPT-4, with the average scores of 15 evaluation indicators exceeding six points. Moreover, the majority of the evaluation dimensions showed only a slight discrepancy in performance compared with the exemplary teaching plans designed by human teachers. These results clearly demonstrate that the teaching plans generated by LLM based on mathematical problem chains excel in dimensions, such as setting instructional objectives, clarifying teaching priorities, designing problem chains, planning teaching activities, knowledge content, and choosing teaching methods and strategies, providing valuable references for teachers' instructional design. Overall, the average score for the overall rating dimension [Avg(F)] was 6.071, and the practical value dimension [Avg(V1)] was 6.085,

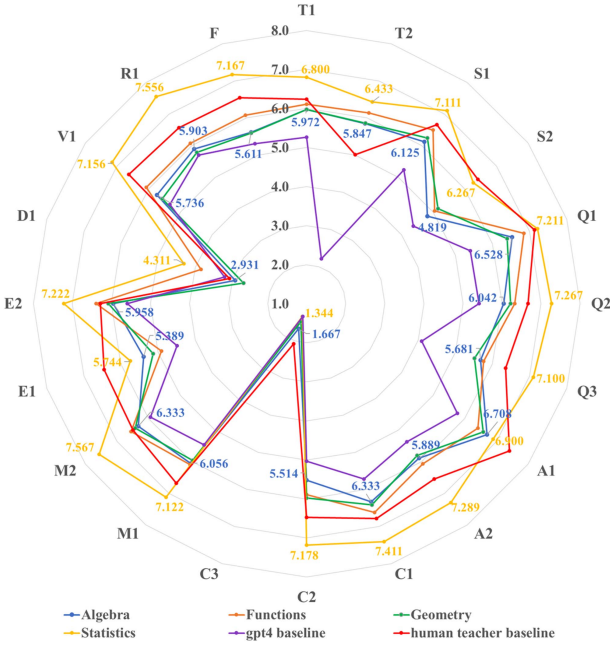


Fig. 3. Average values of each assessment dimension under different knowledge modules.

both of which are higher than the performance in the baseline teaching plan generated by GPT-4. The GPT-4 generated baseline teaching plans performed moderately, with scores typically fluctuating around five points. The evaluation performance of the exemplary teaching plans designed by human teachers was slightly better than those generated by LLMs based on mathematical problem chains. This further indicates that the teaching plans designed and generated using LLM based on mathematical problem chains not only possess good practicality but also can achieve higher usability after review and revision by human teachers.

### B. Cross-Analysis of Dimensions With Knowledge Modules

This section explores the scores of the teaching plans across different knowledge modules and evaluation dimensions. The sample sizes for each module are as follows: “Algebra” module  $N=36$ , “Functions” module  $N=32$ , “Geometry” module  $N=70$ , and “Statistics” module  $N=45$ . The baseline dataset of teaching plans generated by GPT-4 has a sample size of  $N=60$ , and the dataset of exemplary teaching plans designed by human teachers, used as a baseline, also comprises a sample size of  $N=60$  ( $N=15$  for each knowledge module).

By calculating the average scores for each evaluation dimension under different knowledge modules, this study found variances in the average scores between the modules. As shown in Fig. 3, the distribution of average scores across different knowledge modules was depicted, with specific mean values for the “Statistics” and “Algebra” modules highlighted. To explore the differences between various evaluation dimensions and knowledge modules, the Kruskal–Wallis H test (a rank-sum test) was employed. The test results indicated that, except for the C3 dimension, the  $p$ -values for all other dimensions were less than 0.05, signifying significant differences. To determine which

TABLE V  
PAIRWISE COMPARISON OF KNOWLEDGE MODULES ACROSS ASSESSMENT DIMENSIONS

| Dimension | Knowledge modules Avg(SD)       | Pairwise <sup>1</sup> and <sup>2</sup> comparison | H statistic | Adj. $p$ <sup>3</sup> and <sup>4</sup> |
|-----------|---------------------------------|---------------------------------------------------|-------------|----------------------------------------|
| T1        | A: 5.972(0.903) F: 6.110(1.071) | G < S***                                          | -86.945     | 0.000                                  |
|           | G: 5.971(0.921) S: 6.800(0.962) | A < S***                                          | -84.846     | 0.000                                  |
|           |                                 | F < S***                                          | -67.521     | 0.000                                  |
| T2        | A: 5.847(0.883) F: 6.141(0.941) | A < S**                                           | -57.279     | 0.002                                  |
|           | G: 5.864(0.850) S: 6.433(1.102) | G < S***                                          | -55.583     | 0.000                                  |
|           |                                 |                                                   |             |                                        |
| S1        | A: 6.125(1.034) F: 6.500(1.113) | A < S***                                          | -99.781     | 0.000                                  |
|           | G: 6.250(0.930) S: 7.111(0.953) | G < S***                                          | -92.196     | 0.000                                  |
|           |                                 | F < S**                                           | -59.788     | 0.002                                  |
| S2        | A: 4.819(1.755) F: 5.047(1.794) | A < S***                                          | -91.287     | 0.000                                  |
|           | G: 5.150(1.640) S: 6.267(1.490) | F < S***                                          | -75.654     | 0.000                                  |
|           |                                 | G < S***                                          | -72.827     | 0.000                                  |
| Q1        | A: 6.528(1.256) F: 6.844(1.072) | G < S***                                          | -72.242     | 0.000                                  |
|           | G: 6.393(1.142) S: 7.211(0.868) | A < S***                                          | -59.404     | 0.000                                  |
|           |                                 |                                                   |             |                                        |
| Q2        | A: 6.042(1.180) F: 6.328(0.944) | A < S***                                          | -122.324    | 0.000                                  |
|           | G: 6.221(0.906) S: 7.267(0.493) | G < S***                                          | -115.931    | 0.000                                  |
|           |                                 | F < S***                                          | -104.475    | 0.000                                  |
| Q3        | A: 5.681(1.422) F: 5.766(1.445) | G < S***                                          | -111.951    | 0.000                                  |
|           | G: 5.514(1.552) S: 7.100(1.071) | A < S***                                          | -103.560    | 0.000                                  |
|           |                                 | F < S***                                          | -98.866     | 0.000                                  |
| A1        | A: 6.708(1.168) F: 6.422(1.282) | G < S***                                          | -47.449     | 0.026                                  |
|           | G: 6.586(1.399) S: 6.900(1.454) | F < S*                                            |             |                                        |
|           |                                 |                                                   |             |                                        |
| A2        | A: 5.889(1.449) F: 6.063(1.207) | G < S***                                          | -114.227    | 0.000                                  |
|           | G: 5.807(1.419) S: 7.289(0.753) | A < S***                                          | -108.894    | 0.000                                  |
|           |                                 | F < S***                                          | -103.105    | 0.000                                  |
| C1        | A: 6.333(1.061) F: 6.625(0.826) | G < S***                                          | -114.410    | 0.000                                  |
|           | G: 6.414(0.944) S: 7.411(0.616) | A < S***                                          | -111.268    | 0.000                                  |
|           |                                 | F < S***                                          | -92.539     | 0.000                                  |
| C2        | A: 5.514(1.384) F: 5.891(1.170) | A < S***                                          | -138.264    | 0.000                                  |
|           | G: 5.979(1.184) S: 7.178(0.728) | F < S***                                          | -117.122    | 0.000                                  |
|           |                                 | G < S***                                          | -107.587    | 0.000                                  |
| M1        | A: 6.056(1.033) F: 6.094(1.080) | G < S***                                          | -101.807    | 0.000                                  |
|           | G: 5.964(1.102) S: 7.122(1.000) | A < S***                                          | -95.653     | 0.000                                  |
|           |                                 | F < S***                                          | -92.875     | 0.000                                  |
| M2        | A: 6.333(1.007) F: 6.563(1.067) | A < S***                                          | -127.819    | 0.000                                  |
|           | G: 6.421(0.990) S: 7.567(0.637) | G < S***                                          | -119.530    | 0.000                                  |
|           |                                 | F < S***                                          | -104.177    | 0.000                                  |
| E1        | A: 5.389(1.439) F: 4.906(1.875) | G < S***                                          | -49.673     | 0.021                                  |
|           | G: 5.136(1.514) S: 5.744(1.975) | F < S*                                            |             |                                        |
|           |                                 |                                                   |             |                                        |
| E2        | A: 5.958(1.131) F: 6.391(1.002) | A < S***                                          | -122.335    | 0.000                                  |
|           | G: 6.079(0.997) S: 7.222(0.614) | G < S***                                          | -114.496    | 0.000                                  |
|           |                                 | F < S***                                          | -89.080     | 0.000                                  |
| D1        | A: 2.931(1.673) F: 3.844(2.385) | G < S**                                           | -51.464     | 0.001                                  |
|           | G: 2.700(1.984) S: 4.311(3.085) | G < F**                                           | -53.873     | 0.003                                  |
|           |                                 |                                                   |             |                                        |
| V1        | A: 5.736(1.384) F: 6.078(1.103) | G < S***                                          | -118.040    | 0.000                                  |
|           | G: 5.579(1.532) S: 7.156(0.847) | A < S***                                          | -114.892    | 0.000                                  |
|           |                                 | F < S***                                          | -96.182     | 0.000                                  |
| R1        | A: 5.903(1.323) F: 6.078(1.505) | A < S***                                          | -135.169    | 0.000                                  |
|           | G: 5.793(1.765) S: 7.556(1.191) | G < S***                                          | -127.490    | 0.000                                  |
|           |                                 | F < S***                                          | -117.828    | 0.000                                  |
| F         | A: 5.611(1.488) F: 6.078(1.186) | A < S***                                          | -116.842    | 0.000                                  |
|           | G: 5.600(1.604) S: 7.167(0.915) | G < S***                                          | -111.006    | 0.000                                  |
|           |                                 | F < S***                                          | -90.820     | 0.000                                  |

<sup>1</sup> \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , A = Algebra, F = Functions, G = Geometry, S = Statistics.

<sup>2</sup> Testing the null hypothesis that “the distribution of the two knowledge modules is the same.”

<sup>3</sup> Progressive significance is displayed (two-sided test). The significance level is set at 0.05.

<sup>4</sup> Bonferroni correction has been applied to adjust the significance values for multiple tests.

knowledge modules exhibited these significant differences, a posthoc analysis with pairwise comparisons of knowledge modules under each evaluation dimension was conducted (see Table V). Due to the large number of comparisons in the analysis, this study listed only those comparisons that were significant at less than 0.05 after the Bonferroni adjustment.

Looking at the different knowledge modules, in the “Statistics” module, the average scores for most evaluation dimensions were significantly higher than the other three modules. The “Functions,” “Algebra,” and “Geometry” modules had similar average scores across their evaluation dimensions, with fewer significant differences. The evaluation dimensions in the “Functions” module were slightly better on average than the other two

modules. In addition, in the “Statistics” module, the content of the teaching plans scored above 7.0 in 13 evaluation dimensions and above 6.0 in 17 dimensions. For the “Functions” module, four evaluation dimensions had average scores exceeding 6.5, generally fluctuating around 6.0. In the “Geometry” and “Algebra” modules, the average scores for their evaluation dimensions mainly were around 6.0.

Looking at the various evaluation dimensions, the teaching plan contents in the “Statistics” and “Functions” modules performed better in terms of instructional objectives and the identification of teaching priorities and challenges, while other modules faced issues with overly complex expressions of instructional objectives and failure to identify teaching challenges. In the problem chains dimension, each knowledge module demonstrated good quality in introducing problems and linking them cohesively, especially in the Q1 dimension (introduction of problem situations), where the average score generally exceeded 6.5. This indicates that the content designed by LLM can assist teachers in enriching teaching scenarios. However, in designing variation and advanced questions, LLM-generated questions tend to be more challenging and complex compared with textbook questions. In terms of selecting teaching activities, methods, strategies, and evaluation approaches, LLM was able to choose from a variety of activities and evaluation methods provided in the prompt commands, enriching and bolstering the content of the teaching plans. In the A1 dimension (activities that promote teacher–student interaction), the average score generally exceeded 6.5.

However, in the C3 dimension (mathematical culture), the performance of teaching plans across all knowledge modules was poor, with no significant differences observed. Except for the “Statistics” module, the performance in the E1 dimension (identifying individual student differences) was average for the remaining modules. All knowledge modules performed poorly in the D1 dimension (interdisciplinary design). In the R1 dimension (designing teaching content within curriculum standards), only the “Statistics” module scored above 7, outperforming the evaluation performance of the exemplary teaching plans designed by human teachers. However, the other modules and the baseline teaching plans generated by GPT-4 performed poorly. This indicates that the teaching plans generated by LLM tend to either exceed the curriculum scope or fail to cover the required content range for the lesson.

Overall, in terms of overall rating and practical value, the teaching plans generated by GPT-4 based on mathematical problem chains scored above 6, slightly lower than the evaluation performance of the exemplary teaching plans designed by human teachers. This indicates that these teaching plans possess a certain level of utility. The baseline teaching plans generated by GPT-4, however, showed average performance, with scores typically fluctuating between 4 and 5. The teaching plans in the “Statistics” and “Functions” knowledge modules performed well in terms of practicality and usability after modification, demonstrating the potential to assist teachers in instructional design. The “Geometry” module’s plans scored lower, which aligns with the limitation of LLM being language models primarily used for text generation and not yet capable of designing or generating 3-D geometric images.

## V. DISCUSSION

The present study conducted a comprehensive evaluation of the potential of LLMs in instructional design. Building on the results from analyses, and by synthesizing the rationales from the subject matter experts when resolving scoring discrepancies and suggestions from reviewing teachers, this study reached the following observations.

### A. Instructional Design Potential of LLM

Generative AI (Gen-AI) and LLM are not specifically designed for teacher instruction and student learning. Although LLMs are widely used by students in higher education, there is still a lack of research on guiding teachers to utilize LLMs as teaching tools in primary and secondary education [75]. In assessing the potential application of LLM in the field of education, the focus is not only on verifying the accuracy of LLM problem-solving [76] but, more importantly, on examining their performance in specific educational scenarios supported by educational theory. This study views LLM as a “teacher agent focused on instructional design” from the perspective of teacher instructional design, aiming to explore the performance of LLM in instructional design in a manner similar to evaluating human teachers.

In response to research questions 1 and 2, the analysis results indicate that, compared with baseline teaching plans generated by GPT-4 without prompts based on mathematical problem chains, the teaching plans created by GPT-4 with prompts incorporating essential CK, PK, and mathematical problem chains not only achieve seamless integration of teaching phases in problem scenario design but also enable the design of multiple variant problems within the same scenario. This provides references for teachers in reorganizing textbook content and designing connections in problem scenarios. Moreover, LLM performs well in dimensions, such as expressing subject knowledge (CK), setting instructional objectives (PCK), designing problem chains and scenarios, organizing teacher–student interactive activities, and choosing teaching methods and strategies (PK). Furthermore, the quality of teaching plans generated under the “Statistics” and “Functions” knowledge modules is notably high. Some knowledge modules and dimensions even match the performance of the exemplary teaching plans designed by human teachers. This demonstrates the potential of using LLM as a “teacher agent,” capable of assisting real teachers in instructional design, and the possibility of using LLM to help novice teachers enhance the quality and efficiency of their instructional design [58]. However, there is room for improvement in LLM in areas, such as designing interdisciplinary content, integrating historical and cultural elements of mathematics, identifying student learning difficulties, and planning differentiated learning activities.

### B. Assessing the Instructional Design Capabilities of LLM as “Teacher Agent”

Existing research predominantly utilizes tools, such as questionnaires and classroom observation scales, to assess teachers’ instructional design capabilities [32], [34], and some studies

evaluate the effectiveness of preservice teacher training by analyzing teaching plans [36]. The evaluation dimensions in these assessments coincide with the CK and PK components of PCK theory. Researchers have posited that CK is a crucial prerequisite for effective teaching, as a teacher's knowledge impacts the quality of instruction. However, studies also indicate that the simultaneous activation of a teacher's CK and PK fosters the development of PCK [77]. Despite the limitations of PCK theory due to the interrelatedness of its components and the ambiguous conceptual boundaries of these elements, CK and PK constitute the fundamental components of PCK [78]. Discussions on the differences in components of PCK do not affect the use of CK and PK as the primary evaluation dimensions for instructional design in this study.

In addition to incorporating CK and PK as fundamental evaluation dimensions in various elements of instructional design [38], the present study conducted a more comprehensive analysis of the teaching plans generated by LLM. The National Council for the Accreditation of Teacher Education highlights [79] that teaching the history of mathematics is a tool to enrich students' mathematical learning, and teachers should guide students to appreciate the historical and cultural development of the subject knowledge [80]. For teachers aspiring to design interdisciplinary teaching activities but lacking the necessary CK, additional professional training may not timely meet their instructional design needs [81]. LLM, being pretrained on massive datasets, has acquired a rich repository of knowledge. The research leveraged this extensive knowledge base of LLM to generate teaching plan content related to mathematical history stories [40] and interdisciplinary subjects, and included corresponding prompt instructions and evaluation dimensions.

In addition, the design of questions in teaching plans is a vital component reflecting a teacher's instructional design capabilities [82]. Experienced mathematics teachers focus on students' familiarity with practical scenarios, guiding learning through the design of factual, directive, and exploratory questions [83], forming a sequence of problem chains to develop students' analytical, problem-solving, and creativity abilities [84]. However, many novice teachers, when designing questions based on textbooks, often lack experience in integrating real-world problems with mathematical concepts [85], resulting in closed-ended questions presented as isolated learning tasks [86]. This study hopes to use LLM to provide inspiration and reference for teachers in question design. Therefore, prompt instructions and assessment dimensions were devised, both based on mathematical problem chain. Ultimately, an evaluation tool was developed to assess the instructional design capabilities of LLM.

### C. Implications for the Application of LLM in Teacher Instruction

This research combined PCK and instructional design theory, viewing LLM as a "teacher agent focused on instructional design," to explore the integration of Gen-AI into education. Based on the findings of this research, the following implications are proposed.

1) *Implementing Specialized Teacher Training Programs for LLM Prompt Engineering*: As LLM emerges as an innovative tool for assisting instructional design, this study recommends implementing specialized LLM prompt engineering training programs for teachers. These programs should cover key skills [75], such as effectively designing and using prompts to generate diverse instructional activities and classroom scenario presets, adjusting and customizing prompts based on students' learning levels and interests, and integrating school culture, teaching experience, and subject expertise while using LLM to create content that is more engaging and better aligned with instructional objectives. Through this training, teachers will not only act as instructional designers but also as prompt engineers, blending educational practice with technological innovation to create richer and more profound teaching content.

2) *Exploring LLM-Assisted Heuristic Learning Scaffold for Preservice Teacher*: Learning how to restructure instructional content to better align with students' needs presents a challenge for preservice teachers [87]. Preservice teachers tend to favor a textbook-based teaching approach that adheres to a fixed sequence. While this method ensures a basic structure in teaching process, it also constrains their professional development [88]. To break this limitation, some research suggested providing heuristic reflection strategies as scaffolding [89], gradually guiding preservice teachers to assess whether their teaching plans were suitable [90]. If LLMs were incorporated as a form of heuristic content scaffold, once teachers had identified areas for enhancement, LLMs could offer teachers a variety of enhancement options (such as teaching activities for different instructional scenarios and high-quality learning questions [91]), enriching teachers' instructional design experience and insights. This could establish an LLM-assisted heuristic approach for preservice teacher education. This approach could provide preservice teachers with ample inspiration for teaching content design [92], lighten the load of lesson preparation, and alleviate the stress and anxiety associated with how to teach new lesson content. Furthermore, it could also enable teachers to develop more flexible instructional thinking, and enhance the design of teaching activities and the ability to integrate knowledge with context [93].

3) *Establishing a Regulatory Framework for LLM-Generated Educational Content*: Despite the potential value of LLM-designed teaching plans in restructuring instructional content, shortening course durations, and designing personalized learning paths, the complexity of the content generated by LLM can sometimes be too challenging for students with lower ability levels. This diminishes the practicality of LLM as a universally applicable instructional design tool and limits its widespread adoption in nonelite schools [94]. This study recommends the establishment of a dedicated regulatory framework for LLM-generated educational content, which would control the difficulty, depth, and scope of the generated content to ensure its alignment with teaching standards and students' academic levels. The mechanism can help LLM to create course materials that align more closely with curriculum requirements and student capabilities without extensive revisions from instructors. This will enhance the

quality and efficiency of instructional design, making LLM a more universally applicable and effective teaching aid.

4) *Integrating Knowledge Graphs and Specialized LLMs for Mathematics for Customized Instructional Design*: When tasked with generating instructional content that includes 3-D geometric figures, LLM exhibits noticeable limitations. Currently, some LLMs are specifically developed for solving and posing mathematical problems such as those integrated with plugins such as Wolfram Alpha. However, these LLMs' performance in designing instructional activities around mathematical concepts still needs improvement [76] [95]. To overcome these limitations, the present study proposes integrating knowledge graph technology and specialized mathematical LLMs within LLM. These enhancements will help LLM better understand the relationships between knowledge concepts, more accurately identify instructional needs, and design more appropriate teaching content [58] and targeted answer hints [96] for students at different learning stages by providing contextual information about subject knowledge and student learning profiles. This integration will enable the creation of more customized instructional designs.

## VI. RESEARCH LIMITATIONS

This study evaluated the performance of LLM in instructional design, albeit with a few limitations. First, while PCK has been widely acknowledged as a crucial component of teachers' instructional design and classroom teaching abilities, empirical evidence connecting PCK to student learning outcomes remains inconclusive. Therefore, using prompt instructions, teaching plans generated based on the PCK theory still require teachers' review and secondary modifications [97]. Similarly, as a teacher's PCK dynamically grows, the instructional design capabilities of LLMs may also dynamically improve in the future. Second, when constructing the comprehensive evaluation framework based on the PCK theory, this research strived to incorporate all relevant dimensions. However, some dimensions may have been overlooked such as demonstrating a critical understanding of subject knowledge [98]. This study intends to address this in follow-up research to improve the comprehensiveness of the framework further. Finally, GPT-4 was chosen to represent LLMs based on its superior performance across assessment tasks, aiming to explore the potential of the most advanced LLM for instructional design. However, due to limitations in the number of inquiries for the GPT-4 model, this study only evaluated high school mathematics. In the future, it is hoped that the performance and potential of LLMs will be explored across a broader range of subjects.

## VII. CONCLUSION

This study regards LLMs as "teacher agents focused on instructional design" and proposes an evaluation pathway for assessing their instructional design capabilities, similar to the evaluation of human teachers. This pathway includes the creation of instructional designs based on mathematical problem chains, the design of high-quality teaching plan prompts, a dataset of high school mathematics teaching plans generated

TABLE VI  
DESIGN OF PROMPT INSTRUCTIONS CORRESPONDING TO EACH ELEMENT IN THE INSTRUCTIONAL DESIGN FRAMEWORK

| Phase                         | Components of prompt commands                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instructional Objective       | Lesson Name + Knowledge Points + Quantity Limits + Core Disciplinary Literacy + Examples of Instructional Objectives                                                    |
| Teaching Priority             | Instructional Objectives + Lesson Name                                                                                                                                  |
| Teaching Challenge            | Instructional Objectives + Lesson Name                                                                                                                                  |
| Pre-class Introduction        | Lesson Name + Instructional Objectives + Knowledge Points + Methods of Preclass Introduction + Examples of Case Introductions and Problem Chains + Output Format        |
| Explanation of New Knowledge  | Lesson Name + Pre-class Introduction + Knowledge Points + Teaching Activities + Teaching Methods + Examples of Problem Chains and Variant Problems + Output Format      |
| Consolidation and Improvement | Explanation of New Knowledge + Lesson Name + Knowledge Points + Teaching Activities + Teaching Methods + Output Format (Advanced Problems and Advanced Exercises)       |
| Comprehensive Exercise        | Consolidation and Improvement + Lesson Name + Knowledge Points + Teaching Methods + Output Format (Comprehensive Examples and Variant Exercises)                        |
| Lesson Summary                | Preclass Introduction + Explanation of New Knowledge + Consolidation and Improvement + Lesson Name + Knowledge Points + Methods of Lesson Summarization + Output Format |

TABLE VII  
BASIC INFORMATION OF TEACHERS

| Teaching experience | Number | Responsibility                         |
|---------------------|--------|----------------------------------------|
| 5–7 years           | 3      | Scoring                                |
| 11–15 years         | 2      | Balancing major differences in scoring |

by GPT-4, and the tool for assessing LLM instructional design capabilities. The evaluation process systematically examines the quality and potential of LLM instructional designs from multiple assessment dimensions and knowledge modules. This evaluation not only considers the key phases of instructional design but also aims to enhance the richness and practical value of the teaching plan content, thereby enabling a comprehensive assessment of the potential of LLMs in instructional design. As LLMs continue to evolve through ongoing updates and iterations, future research can draw upon this approach to experiment with integrating educational domain knowledge graphs for the automated design of instructional content. This would provide LLMs with both CK and PK, enabling broader evaluations across various subjects. Overall, the results of this study reveal the immense potential and applicational value of LLMs in instructional design, demonstrating exceptional performance across most assessment dimensions and specific knowledge modules. These findings offer valuable references and insights for exploring further applications of LLMs in the field of education, fostering continuous innovation in educational instruction.

## APPENDIX

### DESIGN OF PROMPT INSTRUCTIONS AND BASIC INFORMATION OF TEACHERS

Table VI illustrates the design of prompt instructions corresponding to each instructional phase within the instructional design framework based on a mathematical problem chain. Table VII gives the basic information of the teachers involved in the scoring process.

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